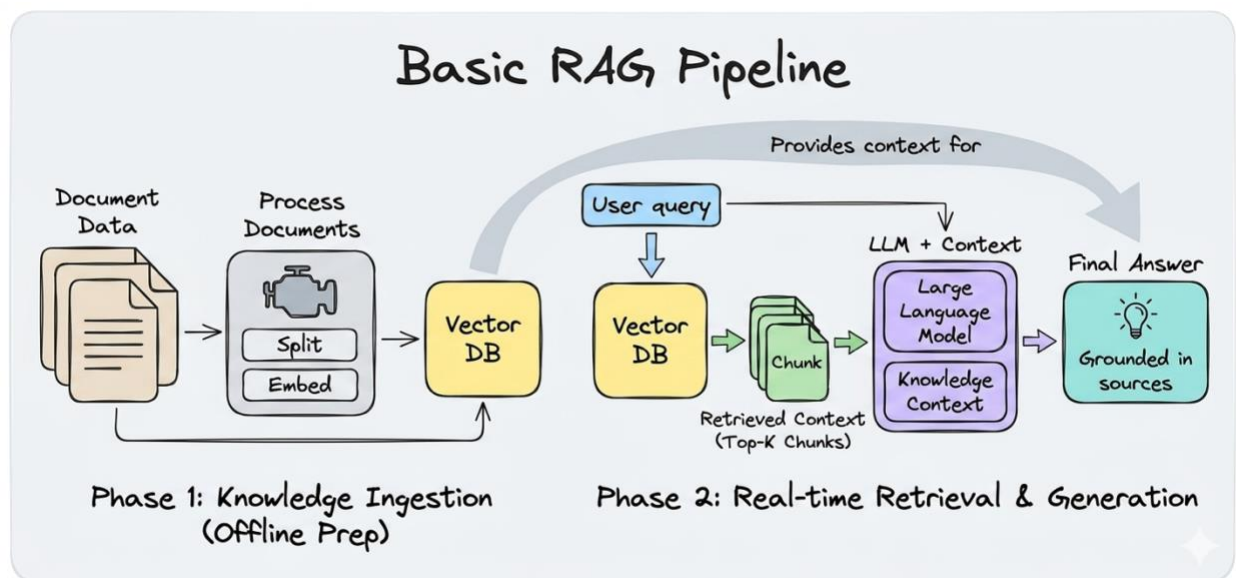


What is RAG?

The full name is **Retrieval-Augmented Generation**. Which sounds impressive but is doing a simple job: it stops an LLM from making things up by forcing it to read your documents before answering.

Here's the problem it solves.

You build a chatbot using ChatGPT or Claude. A user asks “what’s your return policy?” The LLM doesn’t know your return policy — it was trained on the internet, not your business docs. So it either says “I don’t know” or worse, it confidently makes something up (this is called hallucination, and it’s a real problem). RAG fixes this by saying: before the LLM answers, go find the relevant part of the document first, then answer based on that.



Here’s how RAG actually works under the hood. There are two phases:

Phase 1: Indexing (done once, offline)

You take your documents — a PDF, a website, a FAQ file — and you process them:

1. Split them into smaller chunks (500 words or so)
2. Embed each chunk — convert it into a vector, a list of numbers that captures the meaning

3. Store those vectors in a vector database (ChromaDB, Pinecone, etc.)

Phase 2: Querying (every time a user asks something)

1. Take the user's question and embed it — same model, same vector space
2. Search the vector database for chunks whose vectors are closest to the question vector
3. Take the top matching chunks and pass them to the LLM as context
4. The LLM reads those chunks and generates an answer grounded in your actual documents

Folders Based on Pipeline Components

Table 1: Component and it Descriptions

Component	Descriptions
ingestion	handles the input data boundaries (PDFs, Videos, etc.).
pipeline	handles the cognitive core (Chains, Agents, Prompts, Rerankers).
api	handles the application delivery layer (FastAPI).